

- retirement, 176–88, 353–55. *See also* life-cycle, investing over the
- retirement savings puzzle, 187
- return decomposition, factor investing, 445–49
- returns
 - arithmetic versus log returns, 626–29
 - gross returns, 625
- revealed preference, 58
- revisions of past returns, hedge funds, 566
- reward-to-risk ratio, 60
- Rhode Island Retirement Security Act of 2011, 188
- Rhode Island retirement system. *See* Employees' Retirement System of Rhode Island (ERSRI)
- Ricardian equivalence, 477
- riding the yield curve, 458
- Rio Tinto, 49
- risk, preferences for, 37–40
- risk anomaly, 332–44
 - betting-against-beta (BAB) factor, 339–39, 340–41
 - explanations, 341–44
 - low beta anomaly, 335–36
 - low volatility anomaly, 333–34
 - low volatility factor, 306, 339–41
- risk aversion, 40–45, 50–61. *See also* constant relative risk aversion (CRRA) utility
 - as concavity of utility function, 42
 - of market, 58–61, 244–48, 269–70
 - time-varying, 67, 245–46, 436
 - of typical pension fund, 54–58
- risk aversion coefficient, definition, 43
- risk-free bonds. *See* sovereign (risk-free) bonds
- risk-free rate puzzle, 244
- risk parity, special case of mean-variance investing, 101, 103, 104, 106–8, 473
- risk premiums. *See* factor investing
- risk premiums, counter-cyclical, 263–64, 291–92. *See also* predicting returns
- risk and return. *See* performance, risk and return
- risk-return factor analysis, 472
- risk-return trade-off, 222, 269–70
 - affected by investment constraints, 83. *See also* socially responsible investing
- robust estimators, 97–98
- Rochester, University of, 84
- roll return, commodities, 368–70, 458
- Rosen-Roback model, real estate, 374
- Ross, Stephen, 202, 210, 226, 260, 263, 420, 493
- Russia, 141, 247, 415
- sacrifice ratio, 348
- safe assets
 - factors in, 478–80
 - optimal weights, 483–86
 - weighting schemes, 482–83
- safety first utility, 62
- Samsonite, 589
- Samuelson, Paul, 115, 120, 142, 143, 147
- Saudi Arabian Monetary Authority (SAMA), 6, 10
- Save More Tomorrow program, 179
- saving for retirement, 178–79, 187–88
- Scholes, Myron, 115, 388, 560, 585
- Scudder, Stevens & Clark, 504
- SDF. *See* stochastic discount factor (SDF)
- search frictions, illiquid asset markets, 413
- SEC. *See* Securities and Exchange Commission (SEC)
- secondary funds, private equity, 596
- Securities and Exchange Commission (SEC), 394, 417, 520
- Securities Exchange Act of 1934, 561
- selection bias, 377, 424–26, 544
- self-attribution bias, 238, 265
- selling volatility. *See* short volatility
- sensitivity to inputs, mean-variance investing, 96
- separately managed accounts, hedge funds, 566
- September 11, 2001 terrorist attacks, 36, 240
- shareholder activism, 111
- Sharia compliant funds, 109
- Sharpe ratio, definition, 58. *See also* mean-variance investing
- Sharpe, William, 58, 197, 323
- Shiller, Robert, 158, 209, 266, 287
- Shiller inverse price-earnings ratio, 263
- short-sales constraint, 81–82
- short-term investing, long-term investing as, 121–22
- short volatility factor, 35–40, 51, 64–66, 70, 220–22, 339, 445, 453–55, 572–76, 579–80
 - asset allocation with, 53–54, 64–67, 70
 - corporate bonds, 298–300
 - hedge fund returns, 577–78
 - negative price of risk, 220
 - rebalancing as short volatility, 135–41, 146
- side letters, private equity contracts, 612
- Singapore, 12
- size and value-growth benchmarks, 315–18. *See also* factor benchmarks
- size factor, factor risk premiums, 228–30. *See also* SMB factor
- slope factor, bonds, 276
- smart beta, 226, 450. *See also* factor investing
- SMB factor, 226–30, 235–37, 316–19, 324, 327, 340–41
- smoothing bias, illiquid assets, 377
- social factors, in stock market non-participation, 95
- Social Security, 19–20, 23, 175, 185
- socially responsible investing (SRI), 72–73, 110–12
- soft dollar costs, 541
- Solow residuals, 223
- Soros, George, 560
- South Carolina Retirement Systems, 592–93, 614, 616, 619
- South Carolina v. Baker*, 391